

3rdparty Question and libyana answers



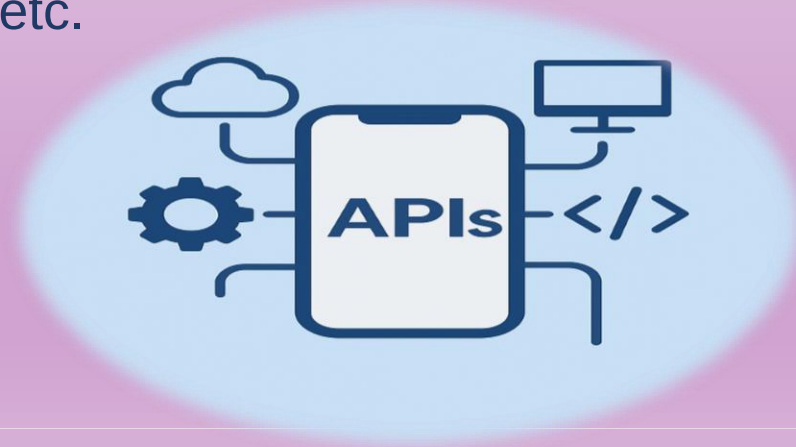
3rdparty Question,

1 -Are there any APIs used at MyLibyana to enhance and ensure internal systems connectivity?

Libyana Reply,

1. MyLibyana APP has been designed with a series of APIs between other related NEs especially BSS core business backend as an effort to achieve different service purposes.

2. These different APIs can be classified into diferent business purposes including Registration/Login Verifications, Customer Profile Inquires, Service Catalog Inquires, Service Order Submissions, Notifications Sending etc.





3rdparty Question,

What security measure are in place to protect APIs' interactions from cyber threats?

Libyana Reply,

Currently, there are 2 types of APIs which have been designed for internet related business interactions, the 1st type is MyLibyana acting as service consumer during APIs interaction, and the 2nd type is MyLibyana acting as service provider during APIs interaction, please kindly refer to details as below, 1.the 1st type while MyLibyana acting as service consumer including business purposes like Logistic Company Interfacing, Digital Online Payment Interfacing, Google Map with main security measures as below,

- a. All the requests and responses messages are going through Libyana firewall which managed by Libyana IT department experts.
- b. Following 3rd party security policies with detailed workflow, e.g. Digital Online Payment Interfacing we have precisely designed the interfaces according to 3rd party protocol requirements with digital handshake process and token verifications.

2. The 2nd type which MyLibyana acting as service provider, including services like User Login Verification, User Service Inquiry, User OTP Request, User Order Submission, and these services will be designed with security measure as below,

- a. All traffic go through Libyana firewall which manage by Libyana IT administrators.
- b. Using HTTPS protocol to ensure all request/response messages are properly encrypted.
- c. Digital Signature to ensure the peer-end identity. d. Token verification to ensure proper authorization.
- e. Vertical Ultra Vires verification. f. Interfaces traffic limitation control



3rdparty Question,

3 -How do APIs contribute to improving service agility and reducing time-to-market for new features?

Libyana Reply,

Current, MyLibyana APIs are based on Whale cloud decades of global best of practices and designed with minimized service duplication and maximized expandability to achieve,

1. Minimized mutual interference between BSS core services and MyLibyana as digital channel, so that new features and services could be designed on MyLibyana more flexibly.

2. Fast rollout and minimized development by reuse existing APIs, e.g. Libyana would like to push an up-sale new offer to users who are both female and with recent 30 days subscription, so that gender could retrieve from customer profile API and purchasing record could get from user historical purchasing record inquiry API.



3rdparty Question,

4 -What are the biggest technical and strategic challenges in API adoption?

Libyana Reply,

series of challenges including Stability, Security, Performance, Expandability, as well as Ease of Use were encountered throughout Whale cloud years of industry practices with corresponding strategies as below,

1. How to maintain API stability?

- a. Simplified and atomic design of single API.
- b. Automated testing workflow with enriched testing scenarios.

2. How to ensure the security of APIs?

- a. Kindly refer to answers of question 2.

3. How to maintain and optimize API performance?

- a. Before commercial use of each API, dedicated performance tests will be held with automated scenarios accordingly.
 - b. After commercial use, APIs will be monitored throughout daily operations, and MS team from whale cloud will periodically collect performance data and suggest with necessary optimization once performance bottlenecks were found.
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3rdparty Question,

5 -What are the primary causes of service outages at MyLibyana, and how are they being addressed?

Libyana Reply,

Currently 2 main causes of MyLibyana service outages are network quality related issues (instability/weak signal) or MyLibyana upgrade with details and suggestions as below,

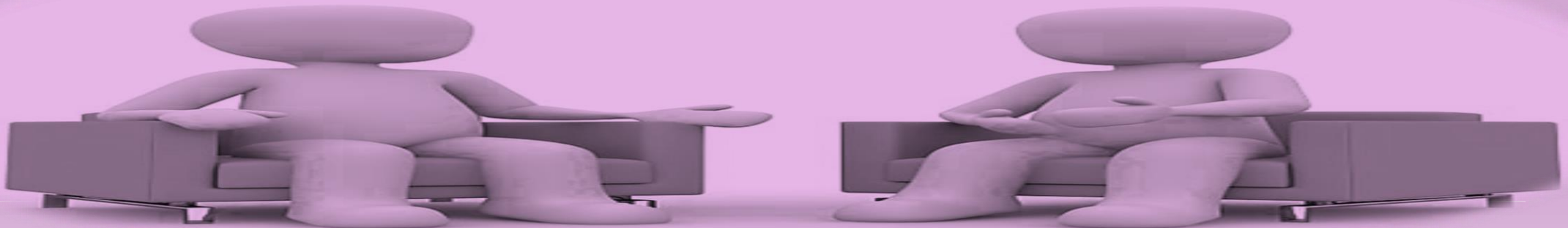
1. For network quality related issues, it is suggested to optimize and upgrade current network so as to gain a relatively higher service stability and continuity, especially geographical hot spot as well as high population density areas.
2. Regarding upgrade operation related service interruption and outages, it is proposed to upgrade current MyLibyana front-end application (Android/IOS) with latest configurability tool set Page Builder + Journey Builder to minimized upgrade caused service outages and maximize daily operation configurabilities

3rdparty Question,

6 - What protocols are in place to minimize downtime and ensure quick service restoration?

Libyana Reply,

1. Based on Whale cloud global practices showing that API protocol actually has very limited impact on service downtime and time cost for service restoration.
2. In addition, as long as MyLibyana is under or back to proper network environment, user service connection will be available automatically, no extra operations from end-user side will be needed





Thanks